

to conclude that the limit of $\varphi_t^{\varepsilon\varepsilon}$ must be the same as that of φ_t^ε , i.e. a Kantorovich potential. Even if the Kantorovich potential is not necessarily unique, its gradient is unique, and we deduce $\nabla \varphi_t^{\varepsilon\varepsilon} \rightarrow \nabla \varphi_t$. $\square$

Note that in the above proof we intensively used the compactness of the domain Ω (and of the torus: the same proof performed with regularization via measures on $\mathbb{R}^d$ seems delicate because of the lack of bounds on $\nabla \varphi_t^{\varepsilon\varepsilon}$). The assumption that the measures are absolutely continuous is used to take advantage of the a.e. convergence of the gradients to a unique limit (the gradient of the limit potential). Without this, one could have problems in recovering full convergence of the sequence.

As a corollary of the previous result we obtain the following useful formula.

Corollary 5.25. *Under the same assumptions of Theorem 5.24, we have,*

$$\frac{d}{dt} \left(\frac{1}{p} W_p^p(\rho_t^{(1)}, \rho_t^{(2)}) \right) = \int_{\Omega} (x - T_t(x)) \cdot \left(\mathbf{v}_t^{(1)}(x) - \mathbf{v}_t^{(2)}(T_t(x)) \right) \rho_t^{(1)}(x) dx,$$

where T_t is the optimal transport map from $\rho_t^{(1)}$ to $\rho_t^{(2)}$ for the cost $\frac{1}{p}|x-y|^p$.

Proof. We just need to re-write the conclusion of Theorem 5.24, using $\nabla \varphi_t(x) = x - T_t(x)$ and $\nabla \psi_t(y) = y - S_t(y)$, where S_t is the optimal transport from $\rho_t^{(2)}$ to $\rho_t^{(1)}$. We use

$$\int (y - S_t(y)) \cdot \mathbf{v}_t^{(2)}(y) d\rho_t^{(2)}(y) = \int (T_t(x) - x) \cdot \mathbf{v}_t^{(2)}(T_t(x)) d\rho_t^{(1)}(x),$$

which is a consequence of $(T_t)_\# \rho_t^{(1)} = \rho_t^{(2)}$ and $S_t = T_t^{-1}$. Hence, the formula given by Theorem 5.24 provides

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{p} W_p^p(\rho_t^{(1)}, \rho_t^{(2)}) \right) \\ = \int_{\Omega} (x - T_t(x)) \cdot \mathbf{v}_t^{(1)}(x) d\rho_t^{(1)}(x) + \int_{\Omega} (T_t(x) - x) \cdot \mathbf{v}_t^{(2)}(T_t(x)) d\rho_t^{(1)}(x), \end{aligned}$$

which gives the claim. $\square$

Some examples of application of the above computation to the uniqueness and the properties of the solution to some evolution PDEs are presented in the Exercise section (**Ex(34)**, **Ex(66)**, **Ex(67)**, **Ex(68)**, **Ex(69)**).

Here, we will only give an example of application to the uniqueness for the continuity equation with given vector field.

Proposition 5.26. *Let $\mathbf{v}_t : \Omega \rightarrow \mathbb{R}^d$ be a family of vector fields on a compact smooth domain Ω , parametrized in time, satisfying a one-sided Lipschitz condition $(\mathbf{v}_t(x) - \mathbf{v}_t(y)) \cdot (x - y) \leq C|x - y|^2$ for a fixed constant C , not depending on x, y nor t . Suppose also that $\mathbf{v}_t \cdot \mathbf{n} = 0$ on $\partial\Omega$ and consider the equation $\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0$ with given initial datum ρ_0 . Then there is at most one solution ρ_t among AC curves in $\mathbb{W}_2(\Omega)$ which are absolutely continuous for a.e. t .*

Proof. Consider two solutions $\rho_t^{(1)}, \rho_t^{(2)}$ and compute the derivative in time of $W_2^2(\rho_t^{(1)}, \rho_t^{(2)})$. Using Corollary 5.25, we have

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{2} W_2^2(\rho_t^{(1)}, \rho_t^{(2)}) \right) &= \int_{\Omega} (x - T(x)) \cdot (\mathbf{v}_t(x) - \mathbf{v}_t(T(x))) \rho_t^{(1)} dx \\ &\leq C \int_{\Omega} |x - T(x)|^2 \rho_t^{(1)} dx = C W_2^2(\rho_t^{(1)}, \rho_t^{(2)}). \end{aligned}$$

This allows to apply Gronwall's lemma and deduce $W_2^2(\rho_t^{(1)}, \rho_t^{(2)}) = 0$ from the fact that the same distance vanishes at $t = 0$. $\square$ $\square$

Note that the above proof could also be used to prove uniqueness in the continuity equation, and applied to Theorem 4.4.

5.4 Constant speed geodesics in $\mathbb{W}_p$

We will see in this section how constant speed geodesics in W_p are related to optimal transport maps. Before doing so, we recall the main facts about geodesics in metric spaces.

Box 5.2. – Good to know! – Constant speed geodesics in general metric spaces

First of all, let us define the length of a curve ω in a general metric space (X, d) .

Definition - For a curve $\omega : [0, 1] \rightarrow X$, let us define

$$\text{Length}(\omega) := \sup \left\{ \sum_{k=0}^{n-1} d(\omega(t_k), \omega(t_{k+1})) : n \geq 1, 0 = t_0 < t_1 < \dots < t_n = 1 \right\}.$$

Note that the same definition could be given for functions defined on $[0, 1]$ and valued in X , not necessarily continuous. The functions ω with $\text{Length}(\omega) < +\infty$ are exactly those which have bounded variation in the sense of the BV functions 1D (see Box 4.4 in Section 4.2.4). It is easy to see that all curves $\omega \in \text{AC}(X)$ satisfy $\text{Length}(\omega) \leq \int_0^1 g(t) dt < +\infty$.

Proposition - For any curve $\omega \in \text{AC}(X)$ we have

$$\text{Length}(\omega) = \int_0^1 |\omega'(t)| dt.$$

We collect now some more definitions.

Definition - A curve $\omega : [0, 1] \rightarrow X$ is said to be a geodesic between x_0 and $x_1 \in X$ if it minimizes the length among all curves such that $\omega(0) = x_0$ and $\omega(1) = x_1$.

A space (X, d) is said to be a *length space* if it holds

$$d(x, y) = \inf \{ \text{Length}(\omega) : \omega \in \text{AC}(X), \omega(0) = x, \omega(1) = y \}.$$

A space (X, d) is said to be a *geodesic space* if it holds

$$d(x, y) = \min \{ \text{Length}(\omega) : \omega \in \text{AC}(X), \omega(0) = x, \omega(1) = y \},$$

i.e. if it is a length space and there exist geodesics between arbitrary points.

We will not enter into details here about the conditions for the existence of geodesics, but the fact that $\text{Length}(\omega)$ is defined as a sup is crucial so to establish semi-continuity results.

Definition - In a length space, a curve $\omega : [0, 1] \rightarrow X$ is said to be a *constant speed geodesic* between $\omega(0)$ and $\omega(1) \in X$ if it satisfies

$$d(\omega(t), \omega(s)) = |t - s|d(\omega(0), \omega(1)) \quad \text{for all } t, s \in [0, 1].$$

It is easy to check that a curve with this property is automatically a geodesic.

The following characterization is useful.

Proposition - Fix an exponent $p > 1$ and consider curves connecting x_0 to x_1 . The three following facts are equivalent

1. ω is a constant speed geodesic,
2. $\omega \in \text{AC}(X)$ and $|\omega'(t)| = d(\omega(0), \omega(1))$ a.e.,
3. ω solves $\min \left\{ \int_0^1 |\omega'(t)|^p dt : \omega(0) = x_0, \omega(1) = x_1 \right\}$.

First, we prove that the space $\mathbb{W}_p(\Omega)$ is a length space, provided Ω is convex.

Theorem 5.27. *Suppose that Ω is convex, take $\mu, \nu \in \mathcal{P}_p(\Omega)$ and $\gamma \in \Pi(\mu, \nu)$ an optimal transport plan for the cost $c(x, y) = |x - y|^p$ ($p \geq 1$). Define $\pi_t : \Omega \times \Omega \rightarrow \Omega$ through $\pi_t(x, y) = (1 - t)x + ty$. Then the curve $\mu_t := (\pi_t)_\# \gamma$ is a constant speed geodesic in $\mathbb{W}_p$ connecting $\mu_0 = \mu$ to $\mu_1 = \nu$.*

In the particular case where μ is absolutely continuous, or in general if $\gamma = \gamma_T$, then this very curve is obtained as $((1 - t)\text{id} + tT)_\# \mu$.

As a consequence, the space $\mathbb{W}_p(\Omega)$ is a geodesic space.

Proof. It is sufficient to prove $W_p(\mu_t, \mu_s) \leq W_p(\mu, \nu)|t - s|$. Indeed, suppose this is proven for every $s > t$: then we would have

$$W_p(\mu, \nu) \leq W_p(\mu, \mu_t) + W_p(\mu_t, \mu_s) + W_p(\mu_s, \nu) \leq W_p(\mu, \nu)(t + (s - t) + (1 - s)) = W_p(\mu, \nu),$$

which implies equality everywhere. To prove the claim, take $\gamma_t^s := (\pi_t, \pi_s)_\# \gamma \in \Pi(\mu_t, \mu_s)$ and compute

$$\begin{aligned} W_p(\mu_t, \mu_s) &\leq \left(\int |x - y|^p d\gamma_t^s \right)^{\frac{1}{p}} = \left(\int |\pi_t(x, y) - \pi_s(x, y)|^p d\gamma \right)^{\frac{1}{p}} \\ &= |t - s| \left(\int |x - y|^p d\gamma \right)^{\frac{1}{p}} = |t - s| W_p(\mu, \nu), \end{aligned}$$

where we used that $|(1 - t)x + ty - (1 - s)x - sy| = |(t - s)(x - y)|$. $\square$ $\square$

Before analyzing the constant speed geodesics in $\mathbb{W}_p$, we want to insist on an important consequence of the fact that we have a geodesic space, and of Theorem 5.14.

The idea behind this result, usually called *Benamou-Brenier formula* and considered as a dynamical version of the Kantorovich problem, will be crucial in Section 6.1 to provide a numerical characterization.

Theorem 5.28. *On a convex and compact domain Ω , given $\mu, \nu \in \mathcal{P}(\Omega)$ we have*

$$W_p^p(\mu, \nu) = \min\{\mathcal{B}_p(\rho, E) : \partial_t \rho + \nabla \cdot E = 0, \rho_0 = \mu, \rho_1 = \nu\},$$

where both ρ and E have to be considered as measures on the space-time $\Omega \times [0, 1]$.

Proof. From the fact that $\mathbb{W}_p$ is a length space we have

$$W_p^p(\mu, \nu) = \left(\min \left\{ \int_0^1 |\rho'(t)| dt, \rho_0 = \mu, \rho_1 = \nu \right\} \right)^p.$$

The minimum can be restricted to constant-speed geodesics and, thanks to the characterization that we gave in Box 5.2, we also have

$$W_p^p(\mu, \nu) = \min \left\{ \int_0^1 |\rho'(t)|^p dt, \rho_0 = \mu, \rho_1 = \nu \right\}.$$

Then we use Theorem 5.14 which allows to replace $|\rho'(t)|$ with $\|\mathbf{v}_t\|_{L^p(\rho_t)}$, thus obtaining

$$W_p^p(\mu, \nu) = \min \left\{ \int_0^1 \|\mathbf{v}_t\|_{L^p(\rho_t)}^p dt, \partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0, \rho_0 = \mu, \rho_1 = \nu \right\}.$$

Setting $E_t = \rho_t \mathbf{v}_t$, the statement is just a re-writing of the above formula. $\square$ $\square$

Coming back to the structure of geodesics in $\mathbb{W}_p$, keeping in mind again the important result of Theorem 5.14, one can wonder what is the velocity field associated to these geodesics. Indeed, the constant-speed geodesic curves μ_t are Lipschitz curves and hence they must admit the existence of a velocity field $\mathbf{v}_t$ (at least if $p > 1$) satisfying the continuity equation $\partial_t \mu_t + \nabla \cdot (\mu_t \mathbf{v}_t) = 0$.

In rough terms this means: take $y \in \text{spt}(\mu_t) \subset \Omega$, for $t \in]0, 1[$, and try to find the speed of the particle(s) passing at y at time t . This should be the value of $\mathbf{v}_t(y)$. It would be easier to answer this question if we had uniqueness of “the particle” passing through y at time t . To provide this uniqueness, we use the following lemma.

Lemma 5.29. *Let Ω be compact and γ be an optimal transport plan for a cost $c(x, y) = h(y - x)$ with h strictly convex, between two probabilities $\mu, \nu \in \mathcal{P}(\Omega)$ and take $t \in]0, 1[$. Define $\mu_t = (\pi_t)_\# \gamma$ with $\pi_t(x, y) = (1 - t)x + ty$, and take $y \in \text{spt}(\mu_t)$. Then there exists a unique pair $(x, z) \in \text{spt}(\gamma)$ such that $y = (1 - t)x + tz$. These values of x and z will be denoted by $X_t(y)$ and $Z_t(y)$, respectively. The two maps X_t and Z_t are also continuous.*

In particular, if $\gamma = \gamma_T$ comes from a transport map, then the map $T_t := (1 - t)\text{id} + tT$ is invertible and $T_t^{-1} = X_t$.

Proof. The claim is essentially the same as in Lemma 4.23, and the uniqueness of $(x, z) \in \text{spt}(\gamma)$ comes from c -cyclical monotonicity of $\text{spt}(\gamma)$. The continuity of X_t and Z_t is obtained by compactness. Take $y_n \rightarrow y$ and suppose (up to subsequences) that $(X_t(y_n), Z_t(y_n)) \rightarrow (X, Z)$. Since $\text{spt}(\gamma)$ is closed then $(X, Z) \in \text{spt}(\gamma)$. Moreover, uniqueness implies $(X, Z) = (X_t(y), Z_t(y))$. Since any limit of converging subsequences must coincide with the value at y , and we work in a compact space, this gives continuity. $\square$

We can now identify the velocity field of the geodesic μ_t : we know that every particle initially located at x moves on a straight line with constant speed $T(x) - x$, which implies $\mathbf{v}_t(y) = (T - \text{id})(T_t^{-1}(y))$. More generally, if γ is not induced by a map, we have $\mathbf{v}_t(y) = Z_t(y) - X_t(y)$.

Proposition 5.30. *Let $\mu_t = (\pi_t)_\# \gamma$ be the geodesic connecting μ to ν introduced above. Then the velocity field $\mathbf{v}_t := Z_t - X_t$ is well defined on $\text{spt}(\mu_t)$ for each $t \in]0, 1[$ and satisfies*

$$\partial_t \mu_t + \nabla \cdot (\mu_t \mathbf{v}_t) = 0, \quad \|\mathbf{v}_t\|_{L^p(\mu_t)} = |\mu'|_p(t) = W_p(\mu, \nu).$$

Proof. We already saw that X_t and Z_t are well-defined, so we only need to check the continuity equation and the L^p estimate. To prove the continuity equation, take $\phi \in C^1$ and compute

$$\begin{aligned} \frac{d}{dt} \int \phi d\mu_t &= \frac{d}{dt} \int \phi((1-t)x + tz) d\gamma(x, z) = \int \nabla \phi((1-t)x + tz) \cdot (z - x) d\gamma(x, z) \\ &= \int \nabla \phi(\pi_t(x, z)) \cdot (Z_t(\pi_t(x, z)) - X_t(\pi_t(x, z))) d\gamma(x, z) \\ &= \int \nabla \phi(y) \cdot (Z_t(y) - X_t(y)) d\mu_t(y). \end{aligned}$$

To compute the L^p norm we have

$$\int |\mathbf{v}_t|^p d\mu_t = \int |Z_t(y) - X_t(y)|^p d\mu_t(y) = \int |z - x|^p d\gamma(x, z) = W_p^p(\mu, \nu),$$

and we used in both the computations the fact that $Z_t(\pi_t(x, z)) - X_t(\pi_t(x, z)) = z - x$ for every $(x, z) \in \text{spt}(\gamma)$. $\square$

We now want to prove that, at least for $p > 1$, all the geodesics for the distance W_p have this form, i.e. they are given by $\mu_t = (\pi_t)_\# \gamma$ for an optimal γ . Non-uniqueness would stay true in case γ is not unique. The proof that we provide here is different from that of [17] and is based on the following representation Theorem (in the spirit of [213]). We use the set of $\text{AC}(\Omega)$ of absolutely continuous curves defined on $[0, 1]$ and valued in Ω , and we denote by $K_p : \text{AC}(\Omega) \rightarrow \mathbb{R} \cup \{+\infty\}$ the p -Kinetic energy, given by $K_p(\omega) := \int_0^1 |\omega'(t)|^p dt$ (which is only finite if ω belongs to the Sobolev space $W^{1,p}([0, 1]) \subset \text{AC}(\Omega)$). As usual, we write $\mathcal{C}$ for $\text{AC}(\Omega)$, for short.

Proposition 5.31. *For every Lipschitz curve $(\mu_t)_t$ in $\mathbb{W}_p(\Omega)$ there exists a measure $Q \in \mathcal{P}(\mathcal{C})$ such that $\mu_t = (e_t)_\# Q$ and $\int_{\mathcal{C}} K_p(\omega) dQ(\omega) \leq \int_0^1 |\mu'|_p(t)^p dt$.*

Proof. First we note that for a.e. t there exists, thanks to Theorem 5.14, a vector field $\mathbf{v}_t \in L^p(\mu_t)$ such that $\partial_t \mu_t + \nabla \cdot (\mathbf{v}_t \mu_t) = 0$, and with $\|\mathbf{v}_t\|_{L^p(\mu_t)} \leq |\mu'| (t)$. The same approximation argument (with the same convolution kernels) as in Theorem 5.14 provides a regularized curve of measures μ_t^ε and a regularized vector field $\mathbf{v}_t^\varepsilon$, with $\|\mathbf{v}_t^\varepsilon\|_{L^p(\mu_t^\varepsilon)} \leq \|\mathbf{v}_t\|_{L^p(\mu_t)}$. This vector field has a flow $Y : \Omega \rightarrow \text{AC}(\Omega)$, where $Y(x)$ is the curve $y_x(\cdot)$, and we know $\mu_t^\varepsilon = (Y_t)_\# \mu_0^\varepsilon$, where $Y_t(x) = Y(x)(t) = y_x(t)$. Let us also define the measure $Q^\varepsilon := Y_\# \mu_0^\varepsilon$. We have $\mu_t^\varepsilon = (e_t)_\# Q^\varepsilon$.

Let us compute $\int_{\mathcal{C}} K_p dQ^\varepsilon$:

$$\begin{aligned} \int_{\mathcal{C}} K_p(\omega) dQ^\varepsilon(\omega) &= \int_{\Omega} \int_0^1 |\mathbf{v}_t^\varepsilon(y_x(t))|^p dt d\mu_0^\varepsilon(x) = \int_0^1 \int_{\Omega} |\mathbf{v}_t^\varepsilon|^p d\mu_t^\varepsilon \\ &\leq \int_0^1 \int_{\Omega} |\mathbf{v}_t|^p d\mu_t = \int_0^1 |\mu'| (t)^p dt. \end{aligned}$$

Now (as we saw many times in Chapter 4), this implies that the measures Q^ε are tight, since there is a bound on $\int_{\mathcal{C}} K_p dQ^\varepsilon$ and K_p is such that

$$\{\omega \in \text{AC}(\Omega) : K_p(\omega) \leq L\}$$

is compact in $C^0([0, 1])$ (for the uniform convergence) for every L . Also, K_p is l.s.c. for the same convergence. Hence, we can obtain the existence of a subsequence such that $Q^\varepsilon \rightharpoonup Q$ and $\int_{\mathcal{C}} K_p dQ \leq \liminf_{\varepsilon} \int_{\mathcal{C}} K_p dQ^\varepsilon$. From $\mu_t^\varepsilon = (e_t)_\# Q^\varepsilon$ we get $\mu_t = (e_t)_\# Q$ and we conclude. $\square$

Proposition 5.32. *Let $(\mu_t)_t$ be a constant speed geodesic in $\mathbb{W}_p(\Omega)$ between μ and ν and suppose $p > 1$. Then there exists an optimal $\gamma \in \Pi(\mu, \nu)$ for the transport cost $c(x, y) = |x - y|^p$ such that for every $t \in [0, 1]$ we have $\mu_t = (\pi_t)_\# \gamma$.*

Proof. Since the curve is a constant speed geodesic, we have $|\mu'| (t) = W_p(\mu, \nu)$ for a.e. t (the metric derivative being computed according to W_p). Applying our previous Proposition 5.31, we get the existence of a measure $Q \in \mathcal{P}(\mathcal{C})$ such that $\mu_t = (e_t)_\# Q$ and $\int_{\mathcal{C}} K_p(\omega) dQ(\omega) \leq \int_0^1 |\mu'| (t)^p dt$.

From Jensen inequality we have $|\omega(0) - \omega(1)|^p \leq K_p(\omega)$, with equality if and only if ω is a segment parametrized with constant speed. Hence, we have

$$\begin{aligned} W_p^p(\mu, \nu) &\leq \int_{\mathcal{C}} |\omega(0) - \omega(1)|^p dQ(\omega) \\ &\leq \int_{\mathcal{C}} K_p(\omega) dQ(\omega) \leq \int_0^1 |\mu'| (t)^p dt \leq W_p^p(\mu, \nu). \end{aligned}$$

The first inequality comes from $\gamma := (e_0, e_1)_\# Q \in \Pi(\mu_0, \mu_1)$; the second is a consequence of the above inequality for K_p ; the last comes from the properties of Q . Thus, all inequalities are equalities. Hence Q is concentrated on curves which are constant speed segments and γ is an optimal transport plan. This proves $\mu_t = (\pi_t)_\# \gamma$. $\square$

Note how the above proof is reminiscent of that of Theorem 4.13, with here a dynamical framework, compared to the static framework in Chapter 4.

It is not difficult to see that both Proposition 5.31 and 5.32 do not hold for $p = 1$. The counter-example is, as usual, given by $\mu_t = (1 - t)\delta_{x_0} + t\delta_{x_1}$.

We finish this section insisting on the role that Wasserstein geodesics play in applications, as they allow to build an interesting interpolation between two distribution of masses (or two histograms), moving slowly the particles from one distribution to the other. For instance, Figure 5.2, shows the interpolation between the chrominance histograms of Figure 2.4. These intermediate histograms could also be used to produce intermediate images.



Figure 5.2: Nine time steps of the interpolation (top, left to right, then bottom, left to right), obtained via a $\mathbb{W}_2$ geodesic curve, between the histograms on the 2D space of chroma signals of the two images of Figure 2.4. Pictures kindly provided by G. Peyré.

5.5 Discussion

5.5.1 The W_∞ distance

The case $p = \infty$ in the definition of W_p deserves special attention. It is easy to guess how to define a W_∞ distance on probability measures:

$$W_\infty(\mu, \nu) := \inf \{ \|x - y\|_{L^\infty(\gamma)} : \gamma \in \Pi(\mu, \nu) \},$$

where the minimization problem in the right-hand side is exactly the one studied in Section 3.2. As we did in the rest of this chapter, the difference here is that, instead of paying attention to the optimal map, we mainly consider the optimal cost as a function of the two measures, and use it as a distance on the space of probabilities.

In order to properly define $W_\infty(\mu, \nu)$, one should restrict to $\mathcal{P}_\infty(\Omega)$, which is the space of measures with bounded support. Otherwise, there could be pairs of measures such that no bounded-displacement plan γ exists. From the results in Section 3.2 we know that, if $\mu \ll \mathcal{L}^d$, then $W_\infty(\mu, \nu)$ is also equal to

$$\inf \{ \|T(x) - x\|_{L^\infty(\mu)} : T_\# \mu = \nu \}.$$